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Separation of Cardiopulmonary Sound Signals for Classification of Respiratory Diseases

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Abstract— **Due to having the mixture of heart and lung sounds, the traditional automatic cardiopulmonary sound signal analysis has some challenges in classification accuracy and robustness. The aim of this study is to improve the performance of respiratory disease classification model by cardiopulmonary sound separation. We used 920 auscultated audio signals involving 127 subjects who were healthy or had different respiratory conditions. For separation of lung sounds from heart sounds and other interferences we have developed an adaptive subspace analysis which iteratively maximizes the kurtosis. The separated lung sound is then classified using random forest classifier. The signals before and after separation are classified by a random forest classifier, and the classification performance is evaluated by cross-validation. The results shows that the classification result of cardiopulmonary sound separation in different age groups (infants and adults) significantly improves.**

Keywords: Cardiopulmonary sound, Lung disease, Singular spectrum analysis, Random forest classification

I. Introduction

Lung disease mainly including chronic obstructive pulmonary disease (COPD), lung cancer and asthma is a major global public health problem. These diseases not only affect the quality of life of patients, but also place an enormous burden on global healthcare systems. Pulmonary sound classification plays an important role in clinical diagnosis, personalized treatment, public health monitoring, medical research and teaching, automation and telemedicine. By improving the accuracy and efficiency of pulmonary sound classification, the management and treatment of respiratory diseases can be significantly improved, and the quality of patients lives can be improved.

Lung sounds are usually collected using a stethoscope, which is also used for heart sound auscultation. The sounds heard by the stethoscope from the chest wall consist mainly of heart sounds, lung sounds, and other underlying sounds, such as thorax and muscle frictions. Heart sounds are mainly caused by the mechanical activity of the heart, including the closing of the heart valves and the vibration of the heart.

A parallel source separation system is proposed to extract heart and lung sounds from single-channel mixed signals based on non-negative matrix decomposition and clustering strategies [1]. In addition, a hard threshold is used in the wavelet transform domain to separate the non-stationary part of the input signal (heart sound) from the stationary part (lung sound). By placing two microphones on the left and right chest walls respectively to collect signals, the fast ICA algorithm was applied to separate the heart sound and lung sound signals [2]. Blind source separation is performed using non-negative matrix factorization (NMF). By constructing spectral mask and using NMF to decompose spectral features, heart and lung sound signals are separated [3]. A new period-encoding depth-using autoencoder (PC-DAE) method is proposed to isolate mixed cardiopulmonary sounds in an unsupervised manner by assuming different periodicity between heart and respiratory rates [4].

Singular spectrum algorithm (SSA) reveals the intrinsic structure of the data by breaking down the time series into interpretable components. Therefore, in this paper, the collected cardiopulmonary sound signals are separated by SSA algorithm in advance to extract purer lung sound signals, so as to improve the accuracy of subsequent classification of lung diseases.

VGG16 model in deep learning has been used to detect and classify lung diseases [5]. The study has a large, publicly available dataset containing X-ray images of COVID-19, pneumonia, and pneumothorax. The method of classifying lung diseases using deep learning and interpretable artificial intelligence (XAI) techniques has been discussed [6]. It mainly uses a variety of deep learning models, including CNN, hybrid model, integrated model and transformer model, and combines XAI technology to improve the interpretability of the model. A multi-class classification method based on convolutional neural networks (CNN) to learn pulmonary disease images has been proposed in [7]. The pre-processing has been performed using center clipping, fine-tuning learning performed using the EfficientNet B7 model, and the Multi GAP structure used to maximize the features of each layer.

Random Forest is an ensemble learning method based on decision trees that is widely used in classification and regression tasks. It constructs multiple decision trees by randomly selecting samples and features during the training process, significantly improving the stability and generalization ability of the model, while being able to assess the importance of features and provide an interpretation of the classification results [8].

In this paper, we propose a method combining singular spectrum algorithm (SSA) and random forest classifier to separate cardiopulmonary sound signals from infants and adults, and classify the extracted lung sounds into various pulmonary diseases. The method is implemented through three main steps: signal separation, feature extraction, and classification including model training and validation.

II. Description of Database

In this study we use the ICBHI 2017 Challenge dataset as the baseline dataset for pulmonary auscultation sounds for the detection of respiratory diseases. It includes 920 recordings ranging in length from 10s-90s from 127 subjects. In total, there are 5.5 hours of recordings, containing 6898 breathing cycles. The data includes both clean breathing sounds and noisy recordings in real life. The data are from either healthy subjects or those having one of the seven categories of respiratory diseases.

III. Methodology

3.1 Singular Spectrum Analysis

SSA is a powerful time series analysis and processing tool, which is often used for single-channel signal separation, noise reduction and pattern recognition [9]. The basic principle is to decompose and reconstruct the one-dimensional time series signal by embedding it into the high-dimensional trajectory space and extracting meaningful subspace structure from it.

A. Embedding

First, the one-dimensional time series is transformed into a high-dimensional trajectory matrix. Given a time series $X = \{x_1, x_2, ..., x_N\}$ and window length L. In generally, L satisfies $2 \le L \le N/2$, In this study, L takes the value of 200. Then construct a trajectory matrix X by the Hankel transform of the signal, where each column is a segment of a time series, defined as:

$$X = \begin{pmatrix} x_1 & x_2 & \cdots & x_K \\ x_2 & x_3 & \cdots & x_{K+1} \\ \cdots & \cdots & \cdots & \cdots \\ x_L & x_{L+1} & \cdots & x_N \end{pmatrix} \tag{1}$$

where the trajectory matrix X is a matrix of L rows and K columns, where each column of X is part of the original signal, L is the window length or embedding dimension, and K is the number of columns of the trajectory matrix, equal to $K = N - L + 1$, where N is the length of the signal.

B. The covariance matrix and eigenvalue decomposit

Next, we calculate the covariance matrix S of the trajectory matrix X. The covariance matrix S is defined as

$$S = X \cdot X^T \tag{2}$$

Then, the EVD is performed and the eigenvalues are sorted and the principal components are extracted for the separation of subsequent signals.

C. Adaptively grouping the desired signal components

To select the desired eigentriples, we multiply a diagonal matrix W by EVD factors and optimize it in each iteration. The W matrix is initialized as an identity matrix, and it is iteratively optimized to maximize the kurtosis of the signal. Kurtosis is a measure of how sharp the signal distribution is. In each iteration, the W matrix is updated by calculating the gradient of kurtosis with respect to W and using gradient descent method. Let the kurtosis function be estimated as follows:

$$Kurt(X) = \frac{\frac{1}{N}\sum_{i=1}^{N}\left(X_i - \mu\right)^4}{\left(\frac{1}{N}\sum_{i=1}^{N}\left(X_i - \mu\right)^2\right)^2} \tag{3}$$

where X is the signal, N is the number of data points, Kurt stands for kurtosis, and μ is the mean of the signal X. In the optimization process, W is then updated as:

$$W(i+1) = W(i) - \alpha \cdot \nabla_w(Kurt) \tag{4}$$

where α is the learning rate and ∇_w is the gradient with respect to W.

D. Reconstruction

According to the optimized W matrix, the covariance matrix Σ is decomposed into the lung sound component Σ_E and the interferences including the heart sound component Σ_I.

Then, the time domain signal is reconstructed by diagonal averaging, transforming the matrix back into a one-dimensional signal while retaining the main characteristics of the desired lung sound signal.

Through these steps, SSA can effectively separate lung sound and heart sound signals, and provide accurate data basis for subsequent signal analysis and diagnosis.

3.2 Kurtosis

Kurtosis is a statistic that measures the peakedness of distribution of a signal's probability distribution, and by maximizing kurtosis, the accuracy of signal separation can be improved, especially when separating non-Gaussian signal components [10]. Therefore, this study proposes to combine SSA with Kurtosis to further optimize the separation effect of heart and lung sounds. The process of combining SSA and Kurtosis algorithm is shown in Fig. 1.

3.3 Random forest model

A random forest is an ensemble learning method that builds multiple decision trees for classification or regression. Each tree is trained on a different sample set and feature subset, and the final prediction is voted on by the results of all trees [8]. The process of building and evaluating a random forest classifier is shown in Fig.1.

For the original dataset, random forest generates multiple training sets through Bootstrap sampling [11]. Each training set is obtained by randomly drawing samples from the original dataset, and the size of each training set is the same as the original data set. For each of the Bootstrap sampling to generate the training set, build a decision tree.

After all the trees are built, the random forest makes a final prediction by integrating the results of all the tree predictions. For classification tasks, the random forest determines the final category through a majority voting mechanism.

IV. Results and Discussion

4.1 Results of separation

According to the characteristics of cardiopulmonary sound data selected in this study, the embedding dimension L of SSA was set to 250 for constructing the trajectory matrix.

The following takes one of the speech signals as an example to illustrate the effectiveness of using SSA and Kurtosis algorithm to separate cardiopulmonary sound signals.

From the signal waveform diagram and the histogram of Fig. 2, we can see that the amplitude of the original signal varies greatly at different time points, and the amplitude value presents a symmetric distribution between -0.2 and 0.2, including a variety of frequency components. The time domain waveform and distribution of separated lung sound signal are roughly similar to the original signal especially when the auscultate heart sound is weak, including a wide frequency range and complex time domain changes. It is consistent with the characteristics of

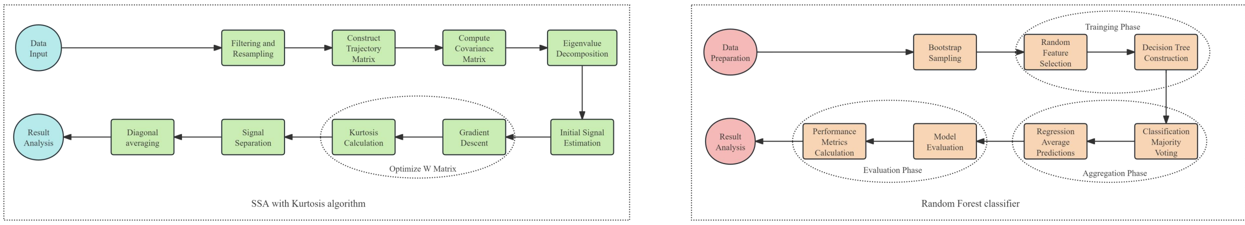

Fig.1 The process of combining SSA and Kurtosis algorithm (left) and the process of building and evaluating a random forest classifier (right).

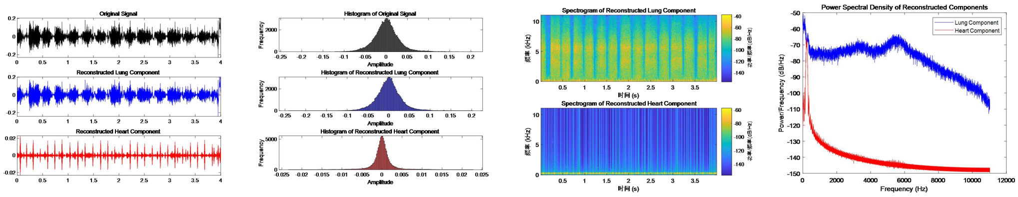

Fig. 2 Time domain waveform of original signal (first of left top), reconstructed lung sound signal (first of left middle) and heart sound signal (first of left bottom) and Histogram of the original signal (second of left top), reconstructed lung sound signal (second of left middle), and heart sound signal (second of left bottom) and Spectrograms of reconstructed lung sound signals (second of right top) and heart sound signals (second of left bottom) and Power spectral density (PSD) of reconstructed lung sound signals and heart sound signals (first of right).

the usual lung sounds which contain high and low frequency components and vary with the respiratory cycle. The time domain waveform of separated heart sound signal is regular, the amplitude is relatively small and the periodicity is strong. The amplitude values of the histogram are concentrated between -0.02 and 0.02, and the distribution is narrow. This is consistent with the physiological characteristics of heart sounds, which are mainly concentrated in the lower frequency range and produce a distinct pulse with each heartbeat.

In Fig. 2, spectrum diagram shows the frequency distribution of signals at different time-frequency points, and power spectral density diagram shows the power distribution of signals at different frequencies. The reconstructed pulmonary sound signal has a significant energy distribution in the frequency range of 0 to 10 kHz, especially with a strong component above 1 kHz, and has a high power over a wide frequency range. The reconstructed heart sound signals are mainly concentrated in the low frequency range (0 to 1 kHz), have low power, and show periodic changes on the timeline. This is consistent with the low-frequency nature of the heart sound signal and the pulses produced by each heartbeat. Spectral diagram and power spectrum density analysis further quantified the distribution of the separated signal in the frequency domain, and proved the efficiency of the separation process.

In addition, in terms of numerical value, the dot product of reconstructed lung sound signal and heart sound signal is 1.480311, and Pearson correlation coefficient is 0.172788, indicating that the linear correlation between the two signals is weak, and there is a certain independence in the time domain.

4.2 Results of lung diseases classification

The Table 1 and Table 2 illustrates the performance of the classifier performance before and after applying SSA with balanced dataset.

Precision represents the percentage of all samples correctly classified. Recall represents the proportion of all samples. F1 Score is the harmonic average of accuracy rate and recall rate, and is a comprehensive evaluation of the two.

In general, the classification model of cardiopulmonary sound after separation has different performance in different categories.

For infant data, the classification performance of the bronchitis and upper respiratory infection categories improved significantly. However, for the healthy and pneumonia categories, the model performance did not significantly improved or deteriorate. When the adult data were added, the performance of the model in the identification of bronchitis, healthy and

Table 1: The classification performance before and after lung sound separation (infants).

Disease Category	**Precision (After)**	Precision (Before)	**Recall (After)**	Recall (Before)	**F1 Score (After)**	F1 Score (Before)
Bronchiolitis	**0.66667**	NAN	**0.66667**	0	**0.66667**	NAN
Healthy	**0**	0	**0**	0	**NAN**	NAN
Pneumonia	**0.33333**	0.5	**0.66667**	1	**0.44444**	0.66667
URTI	**1**	0.25	**0.66667**	0.3333	**0.8**	0.28571

Table 2: The classification performance before and after lung sound separation (infants and adults).

Disease Category	**Precision (After)**	Precision (Before)	**Recall (After)**	Recall (Before)	**F1 Score (After)**	F1 Score (Before)
Bronchiectasis	**0.75**	0.73333	**0.92308**	0.84615	**0.82759**	0.78571
Bronchiolitis	**0.41667**	0.4375	**0.38462**	0.53846	**0.4**	0.48276
COPD	**0.92308**	0.92308	**0.92308**	0.92308	**0.92308**	0.92308
Healthy	**0.4**	0.36364	**0.30769**	0.30769	**0.34783**	0.33333
Pneumonia	**0.75**	0.66667	**0.92308**	0.76923	**0.82759**	0.71429
URTI	**0.36364**	0.125	**0.30769**	0.076923	**0.33333**	0.095238

pneumonia was different, and the overall performance of the identification of pneumonia and upper respiratory tract infection was significantly improved.

Classification of the separated single infant samples and the addition of the adult samples were generally better than classification of the unseparated samples. After separation, the model has higher accuracy and lower loss in training and validation data. The unseparated models showed obvious signs of overfitting, the verification accuracy and loss fluctuated greatly, and the classification performance was poor.

The overall classification accuracy with balanced dataset of infant samples before and after separation was 0.3333 and 0.5, and that of all samples before and after separation was 0.57692 and 0.62821, and the accuracy with unbalanced dataset of all samples before and after separation was 0.91385 and 0.91603, respectively. In conclusion, the separation of cardiopulmonary sounds is helpful to improve the accuracy and stability of respiratory disease classification.

V. Conclusions

In this study, SSA combined with Kurtosis optimization is used to improve the sound signal separation. The performance of the classification model before and after separation is evaluated by using a random forest classifier and cross-validation method, and a complete evaluation process and result analysis method are provided. The separation of cardiopulmonary sound signals from infants and adults significantly improved the performance to respiratory disease classification model, and provided empirical support for the application of cardiopulmonary sound separation technology.

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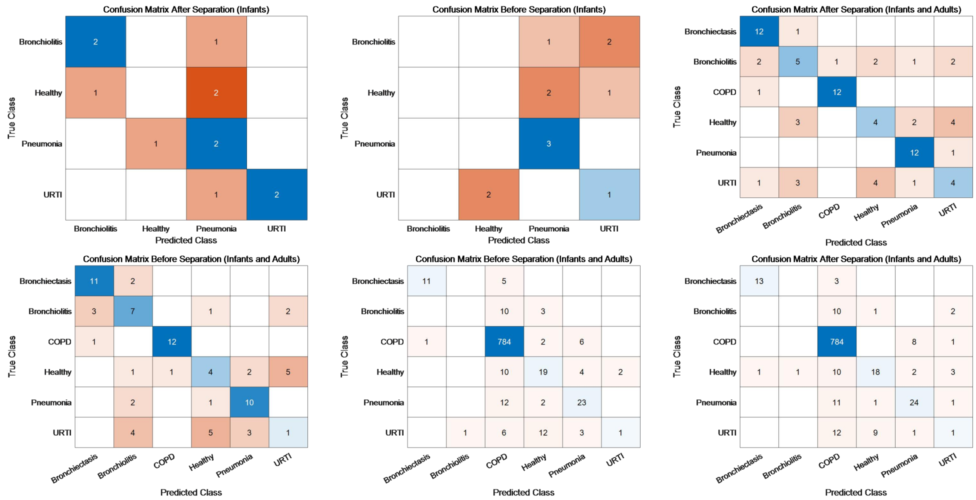

Fig. 3 Confusion Matrix After (left of top) and Before (middle of top) Separation of Infants and After (right of top) and Before (left of bottom) Separation of Infants and Adults with Balanced Dataset and After (middle of bottom) and Before (right of bottom) Separation of Infants and Adults with Unbalanced Dataset.

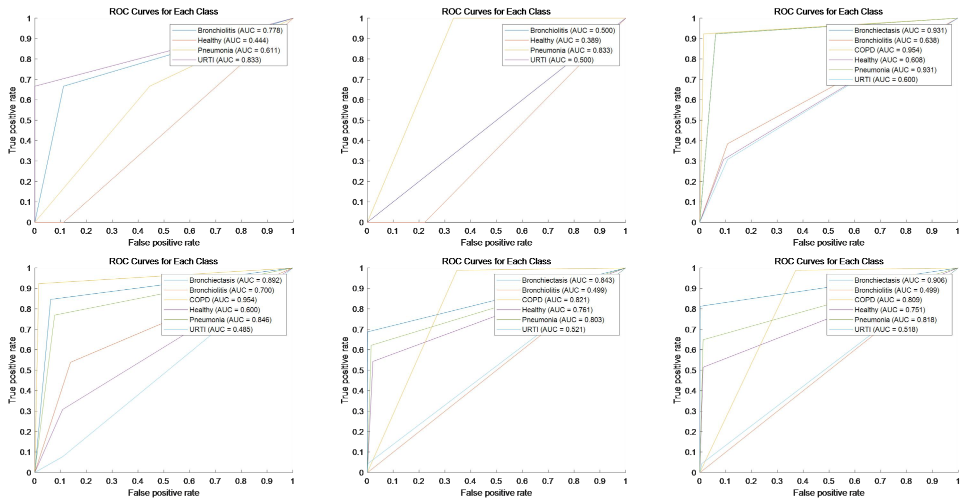

Fig. 4 ROC Curves After (left of top) and Before (middle of top) Separation of Infants and After (right of top) and Before (left of bottom) Separation of Infants and Adults with Balanced Dataset and After (middle of bottom) and Before (right of bottom) Separation of Infants and Adults with Unbalanced Dataset.

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